3D Views for Medical Reports

Software Requirements Document · CISC 3171: Introduction to Software Engineering

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1. Purpose

3D Views for Medical Reports is a web app that shows patients where their bones are broken. Imaging reports are written by radiologists for other doctors, and patients often read them in a patient portal before anyone has explained them. A phrase like "nondisplaced fracture of the distal left radius" tells most people very little. The patient adds their report, the app picks out each fracture it describes, and a 3D skeleton shows each one on the right bone and the right side. As follow-up reports are added, the patient can step through their recovery and compare any two visits. The skeleton is a standard model, not the patient's own bones, and the app never diagnoses, predicts healing, or gives advice.

2. Scope

2.1 In scope

- Adding an imaging report as a PDF, as pasted text, or from a picture of the report (a screenshot, a scanned PDF, or a photo of a printed copy)
- Reports on the bones of the arms and legs, including the shoulders, pelvis, hands, and feet (Appendix A.3)
- Picking out each fracture a report describes, with the patient confirming or correcting the list before it is shown
- A 3D model of an adult skeleton with every confirmed fracture marked on the correct bone, side, and part of the bone (Appendix C.1)
- The original report, unchanged, available beside the model
- A patient's reports in exam-date order, and comparing any two of them
- Attaching the patient's own scan images to a report and viewing them beside the model
- Explanations of fracture terms, written in advance by the team
- Accounts that keep a patient's reports between visits and across computers
- Trying the app with one report before creating an account
- Read-only sharing of chosen reports with non-users (family members or caretakers)

2.2 Out of scope

These are not part of this product now or soon. Capabilities we may build next are listed only in Future Directions (Section 8).

- Building a 3D model of the patient's bones from X-ray images, or from photos or screenshots of any scan. These images have no depth (Appendix A.2).
- Detecting fractures in scan images. Only the written report is read for findings.

- Predicting how or when a fracture will heal
- Diagnosis, treatment advice, or any statement the report does not make
- Summaries of a report, or any medical text written by software while the app runs (NFR6)
- Findings other than fractures, such as dislocations, arthritis, or ligament tears. They stay in the original report, unmarked.
- Reading reports written in languages other than English
- Healthcare professionals as users

3. Context

3.1 User types

Patient. The person the reports are about, or someone managing reports on their behalf, such as a parent. A patient has an account, adds reports, checks the findings, and decides who else can see them. The app does not check whose reports are in an account, so a parent managing a child's reports has exactly the same capabilities as an adult managing their own.

Guest. Someone trying the app before creating an account. A guest can add one report and see it on the model, but nothing is kept: the report is deleted within 24 hours. A guest cannot use the timeline or share.

Viewer. A family member or caretaker who opens the shared report of a patient. A viewer needs no account, sees only the reports in that link (or that are explicitly shared with them), and cannot add, change, or share anything. Their access ends when the patient revokes the link or their direct account access.

Capability	Patient	Guest	Viewer
Add reports	Yes	One, deleted within 24 hours	No
Check and correct findings	Yes	Yes	No
See the model, original report, and scan images	Their own reports	Their one report, without scan images	Only the reports in the link or directly shared
Use the timeline and compare reports	Yes	No	Only the reports in the link or directly shared
Share reports and revoke links/access	Yes	No	No
Needs an account	Yes	No	No

3.2 Where the model's information comes from

The written report is the only source of what the model shows. Scan images are there for the patient to compare against, and never move a marker.

Source	What the app takes from it	What it is never used for
The written report	Each fracture's bone, side, part of the bone, type words, certainty, and healing words; the exam date; the views taken	Anything the report does not state
The patient	Corrections to the findings, a finding the app missed, the exam date when the report has none, and which view each scan image shows	Changing the original report
Scan images	Nothing. They are shown beside the model, turned to the same view.	Placing, moving, or removing a marker

A single X-ray image has no depth, so it cannot be turned into an accurate 3D model of the patient's own bone (Appendix A.2). The report, by contrast, names the bone, the side, and the part of the bone in words the app can match.

3.3 Operating environment

- A laptop or desktop computer running a current version of Chrome, Safari, Firefox, or Edge, with nothing to install
- An internet connection whenever the app is in use. Reports are kept online so the patient can reach them from any computer.
- The patient's own copy of the report, from their doctor, imaging center, or patient portal

3.4 Assumptions

- Patients can get a copy of their imaging report as a PDF, as text on a patient-portal page, or as a printout.
- Reports are written in English, in standard radiology language that names the bone and, for an arm or leg, the side, either in the finding or in the Exam line (Appendix A.1).
- Each report describes one imaging exam and states its date. When it doesn't, the patient knows the date.
- A free-to-use 3D model of an adult skeleton exists in which each of the 126 covered bones is a separate, named part.
- Patients add only their own reports, or reports of someone they manage with that person's permission.
- Scan images are pictures patients already have: images exported from a portal or viewer, or photos of a screen or film. The app does not produce them.
- Patients have the right to share their own reports with anyone they choose.

4. Key Features

The 3D skeleton (F3) is the center of the product. The other P1 features get a report into it, check what it shows, and keep it over time. P1 means the product does not exist without it; P2 means we ship without it if we run out of time.

ID	Feature	What it covers	Priority	Stories
F1	Adding a Report	Adding a PDF or pasted text, dating it by its exam, deleting it	P1	3
F2	Checking the Findings	Picking out each fracture, "no" and "possible" fractures, confirming and correcting, findings that can't be placed	P1	4
F3	The 3D Skeleton	Fractures marked on the right bone, side, and part; turning and zooming; going to each fracture	P1	3
F4	Report and Model Together	The original report beside the model; going between report wording and markers	P1	2
F5	Recovery Timeline	Reports in exam order, comparing two, healing status	P1	3
F6	Accounts	Creating an account, signing in from any computer, deleting the account	P1	3
F7	Scan Images	Attaching the patient's own scan images; turning the model to match an image's view	P2	2
F8	Pictures of Reports	Adding a report from a screenshot, scanned PDF, or photo	P2	1
F9	Understanding the Terms	Pre-written explanations of fracture terms, in findings and in the report	P2	2
F10	Sharing	Read-only sharing through temporary links or invitations to registered users; saving shared reports in the viewer's account; viewing and revoking access	P2	3
F11	Trying Without an Account	One report as a guest; keeping it by creating an account	P2	2

P1 holds 18 stories and P2 holds 10. If we fall behind, P2 features are cut from the bottom of the table up, starting with F11.

5. User Stories

F1: Adding a Report

S1.1: Adding a report

As a patient, I want to add my imaging report, so the app can show me what it describes.

Acceptance criteria

- A report can be added as a PDF file or as pasted text
- A pasted report is treated exactly like a PDF report from then on
- A PDF report can be up to 10 MB and 10 pages long

Scenario 1: Unsupported document file

Given the file is not a PDF

When I attempt to add it as a PDF report

Then the app informs me that the file format is not supported, and nothing is saved.

Scenario 2: File exceeds the limit

Given I select a PDF file larger than 10 MB

When I attempt to add it

Then the app informs me that the file exceeds the maximum allowed size, and nothing is saved.

Scenario 3: Page Count Exceeds the Limit

Given I select a PDF file containing more than 10 pages

When I attempt to add it

Then the app informs me that the file exceeds the maximum allowed page count, and nothing is saved.

Scenario 4: No readable text

Given the PDF has no text the app can read, such as a scanned report

When I attempt to add it

Then the app informs me that it could not find any text in the file and that I can paste the report text instead, and nothing is saved.

S1.2: Dating a report by its exam

As a patient, I want each report dated by the day of the exam, so my reports line up in the order my injury actually progressed.

Acceptance criteria

- A report cannot be saved without an exam date, and the date cannot be in the future
- Dates in the report's Comparison section are never used as its exam date (Appendix A.1)
- I can change a report's exam date at any time

Given the report states one exam date
When I add it
Then that date is filled in for me, and I can change it before saving

Given the report states no exam date, or several dates the app cannot choose between
When I add it
Then I enter the date myself before the report is saved

S1.3: Deleting a report

As a patient, I want to delete a report, so a wrong file or a report I no longer want isn't kept.

Acceptance criteria

- I am asked to confirm before a report is deleted
- Deleting a report also deletes its findings and scan images, and removes it from my timeline and from every share link that included it or user I've given access to
- A deleted report cannot be recovered

F2: Checking the Findings

S2.1: Picking out each fracture

As a patient, I want the app to pick out each fracture my report describes, so I don't have to work out which sentences matter.

Acceptance criteria

- Each fracture the report describes becomes exactly one finding, even when it is described in both the Findings and the Impression (Appendix A.1)
- Each finding records the bone, the side, and the part of the bone when the report states them, plus every fracture-type, certainty, and healing word used for it (Appendix A.4 to A.6)
- When a finding doesn't state a side, it takes the side named in the Exam line, if that line names exactly one side
- A finding on a hand or foot records which finger or toe, whether the report names it or numbers it (Appendix A.4)
- Named fractures, such as "Colles fracture" or "boxer's fracture", are recorded on the bone and part they refer to (Appendix A.5)
- Each finding keeps the exact report wording it came from

S2.2: Reading "no fracture" and "possible fracture"

As a patient, I want negative and uncertain wording read correctly, so I'm never shown a fracture my report rules out.

Acceptance criteria

Given the report states that a fracture is absent, such as "no fracture of the ulna" or "negative for fracture"
When the report is read
Then no finding is created for it

Given the report describes a fracture as possible but not certain, such as "suspected" or "cannot be excluded"
When the report is read
Then the finding is created and marked as uncertain

Given the report describes no fractures at all
When the report is read
Then I am told no fractures were found, and the model shows no markers

S2.3: Confirming the findings

As a patient, I want to check what the app found before I see the model, so a reading mistake doesn't put a fracture on the wrong bone.

Acceptance criteria

- Before a report is first shown on the model, I see every finding taken from it, each with the report wording it came from
- I can remove a finding, or change its bone, side, or part of the bone
- I can add a finding the app missed by choosing its bone, side, and part from the bones the app covers
- A finding I added or changed is labeled as entered by me wherever it appears
- I can return to a report's findings and change them at any time

S2.4: Findings the app can't place

As a patient, I want to be told when a fracture can't be shown on the model, so I never mistake a guess for what my report says.

Acceptance criteria

Given a finding is on a bone the app doesn't cover, such as a rib or a vertebra
When the report is shown
Then the finding is listed as outside the bones the app covers, and no marker is drawn

Given a finding is on an arm or leg bone, and neither the finding nor the Exam line says which side
When the report is shown
Then the finding is listed as "side not stated", and no marker is drawn until I set the side (S2.3)

Given a finding names a region, such as "wrist" or "ankle", but not a bone
When the report is shown
Then the finding is listed as "bone not stated", and no marker is drawn until I choose the bone (S2.3)

F3: The 3D Skeleton

S3.1: Seeing my fractures on the skeleton

As a patient, I want each of my fractures marked on a 3D skeleton, so I can see where I'm hurt instead of decoding the report.

Acceptance criteria

- The model is a complete adult skeleton in the standard anatomical position (Appendix A.4)
- Each placed finding has exactly one marker, on the bone and side the finding names
- Left and right are the patient's own left and right, never the viewer's (Appendix A.4)
- When a finding names the part of the bone, the marker is on that part; when it doesn't, the whole bone is marked
- Uncertain findings (S2.2) look different from certain ones
- Each marker identifies its bone by both its medical name and its everyday name, such as "radius (forearm bone, thumb side)"

S3.2: Moving around the model

As a patient, I want to turn and zoom the skeleton, so I can look at a fracture from whatever angle makes sense to me.

Acceptance criteria

- I can rotate the model in any direction, zoom in and out, and return to the starting view
- I can go straight to a front, back, left-side, or right-side view
- I can zoom in far enough to tell apart the bones of a single finger or toe
- Markers stay on their bones from every angle and at every zoom level

S3.3: Going to each fracture

As a patient, I want to go straight to each fracture, so I don't have to hunt for small bones on a full skeleton.

Acceptance criteria

- The report's findings are listed while the model is in view
- Choosing a finding turns and zooms the model so that fracture is in the center of the view
- I can step through the findings one after another
- I can show an injured bone on its own, without the bones around it, and then bring back the full skeleton

F4: Report and Model Together

S4.1: The original report beside the model

As a patient, I want my original report next to the model, so I can check what the app shows against what my radiologist wrote.

Acceptance criteria

- The original report and the model can be seen at the same time
- Every page of the report can be read while the model stays in view
- The model can also be viewed on its own, and the report brought back

S4.2: Going between the words and the markers

As a patient, I want to go from a sentence in my report to the marker it produced, and back, so I can connect the medical wording to a place on my body.

Acceptance criteria

- Report wording that produced a finding is marked as such in the report
- Selecting that wording brings its marker into view on the model
- Selecting a marker brings the wording it came from into view in the report
- A finding I entered myself (S2.3) has no wording to show and is labeled as entered by me

F5: Recovery Timeline

S5.1: My reports in order

As a patient, I want my reports in the order the exams happened, so I can follow my injury from the first visit to the latest.

Acceptance criteria

- My reports are listed by exam date, oldest first, each with its date and the bones it involves
- A report added later with an earlier exam date takes its place by exam date, not by when it was added
- Reports with the same exam date keep the order in which I added them
- I can move to the next older or newer report, and the model changes to show that report's findings

S5.2: Comparing two reports

As a patient, I want to compare any two of my reports, so I can see what changed between visits.

Acceptance criteria

- I can choose any two of my reports and see both reports' findings at the same time
- Turning or zooming during a comparison keeps both reports seen from the same angle
- For each finding, I can tell whether it appears in both reports, only the earlier one, or only the later one

- A finding in the later report counts as the same injury as one in the earlier report when both are on the same bone and side (OQ3)

S5.3: How each fracture is healing

As a patient, I want each fracture to show how my report describes its healing, so I can see progress at a glance.

Acceptance criteria

- The app never works out a healing status from the time that has passed, or from anything other than the report's words

Given a report describes a fracture with healing words, such as "callus formation" or "healed"
When that report is shown on the model
Then the fracture's marker shows that healing status, in the report's own words (Appendix A.6)

Given a report describes a fracture with no healing words
When that report is shown on the model
Then no healing status is shown for that fracture

F6: Accounts

S6.1: Creating an account

As a patient, I want an account, so my reports are kept between visits and my timeline can grow over weeks.

Acceptance criteria

- Creating an account requires an email address I confirm is mine, and nothing else about me
- One email address can belong to only one account
- An account cannot store reports until its email address is confirmed

S6.2: Signing in from any computer

As a patient, I want to sign in from any computer, so my reports are with me at home, at work, or at the doctor's office.

Acceptance criteria

- After I sign out, nothing of mine can be seen on that computer
- If I can't sign in, I can regain access through my confirmed email address

Given I added reports while signed in on one computer
When I sign in on another
Then every report, finding, and scan image I added is there

S6.3: Deleting my account

As a patient, I want to delete my account and everything in it, so my medical information doesn't stay online after I stop using the app.

Acceptance criteria

- I must sign in again to confirm before my account is deleted
- Deleting my account deletes every report, including its findings, scan images, share links, and access.
- Once the account is deleted, its email address can be used to create a new, empty account

F7: Scan Images

S7.1: Attaching scan images

As a patient, I want to attach images of my actual scan to a report, so I can look at my real X-ray and the model together.

Acceptance criteria

- I can attach up to 10 images to a report, each a JPEG, PNG, or PDF of up to 20 MB
- An attached image and the model can be seen at the same time
- Attached images are shown exactly as I added them, and I can remove any of them
- Attaching or removing an image never adds, moves, or removes a marker

S7.2: Matching the model to an image's view

As a patient, I want the model turned to the angle my X-ray was taken from, so the bone on the model looks like the bone in my picture.

Acceptance criteria

- I can label each attached image with the view it shows, choosing from the views the report names, or from front, side, and angled when it names none (Appendix A.7)
- When I look at a labeled image, the model turns to show the injured bone from that same view
- When I look at an unlabeled image, the model stays where it is

S7.3: Viewing report and sort by Date or Bone

As a patient, I want to organize and view my medical reports and scans by date or by bone, so I can easily find records from a specific appointment or track the recovery of a particular bone.

Acceptance criteria

- Users can choose between two viewing option: by Date / by Bones
- By date display all reports in chronological order based on the examination date.
- By Bones allows the user to select a bone or fracture marker on the 3D model and displays all reports and attached scans associated with that bone.
- The user can switch between the two viewing options at any time.
- If no reports are associated with the selected bone, the app displays a message indicating that no related records were found.

Scenario 1: Viewing Reports by Date

Given I have multiple medical reports saved in my account

When I select "By Date"

Then it will display all my reports in chronological order based on their exam dates.

Scenario 2: Viewing Reports by Bone

Given I have medical reports associated with different bones

When I select "By Bone" and click a bone or fracture marker on the 3D model

Then it will display all reports and attached scans associated with the selected bone, organized by exam date.

Scenario 3: Switching Between Views

Given I am viewing reports organized by date or bone

When I switch to the other viewing option

Then it will update the displayed reports according to my selection.

Scenario 4: No Related Reports

Given I select a bone that has no associated reports

When the app filters my records by that bone

Then the app displays a message indicating that no related reports were found.

F8: Pictures of Reports

S8.1: Adding a report from a picture

As a patient, I want to add a report from a screenshot, a scanned PDF, or a photo of a printed copy, so I can use a report I can't download as text.

Acceptance criteria

- A report can be added from up to 10 JPEG or PNG images, or from a PDF of scanned pages, in the page order I choose
- Before findings are picked out, I see the text the app read from the pictures, and I can correct it
- The pictures themselves are kept as the original report
- For these files, this replaces the "no text the app can read" refusal in S1.1

Scenario 1: Unreadable

Given a page is too blurry, dark, or cut off for its text to be read
When I add it
Then I am told which page could not be read and can replace it, and nothing is saved until every page can be read

Scenario 2: Unreadable Report Text (See OQ2)

Given I upload a scanned PDF or image containing text that can't be read

When I add it

Then I am told that the report text could not be read and can replace the file or image, and nothing is saved until readable text is provided.

F9: Understanding the Terms

S9.1: Explaining the words in my findings

As a patient, I want the medical words in my findings explained in everyday language, so I understand what my report is telling me.

Acceptance criteria

- Every word in a finding that is on the app's term list has an explanation I can see without leaving the model
- Each explanation is at most two sentences and scores at or below US grade 8 on the Flesch-Kincaid grade level
- Words not on the term list are shown as written, with no explanation

S9.2: Explaining terms in the original report

As a patient, I want the same explanations wherever a term appears in my original report, so I can understand the whole report and not only its findings.

Acceptance criteria

- Every term on the term list is marked wherever it appears in the original report
- I can see a marked term's explanation without leaving the report

F10: Sharing

S10.1: Sharing reports

As a patient, I want to share chosen reports with a family member or caretaker, so they understand my injury without me explaining it from memory.

Acceptance criteria

- I choose which of my reports a share link (or direct access) covers, and I can create several links covering different reports
- Before a link is created, I am told that anyone who has it can view those reports
- By default, a link stops working 30 days after it is created
    - Users can customize their default link expiration (e.g., from 30 days to 14) and choose the expiration for each link
    - The same logic applies for direct access (account → account sharing, without links)

S10.2: Seeing and revoking links/access

As a patient, I want to see every link I've shared or users with direct access to my reports, with the ability to revoke them, so I can manage who sees my medical information.

Acceptance criteria

- For a report, I can see a list of users with direct access, when they were given access, and when they last opened it.
- I can see each active link, the reports it covers, when it was created, and when it was last opened
    - This includes permanent links (shared reports)

Given I revoke a link or access
When anyone opens it afterward
Then they are told it is no longer available, and see nothing of the report (including copies they may have created)

S10.3: Viewing shared reports

As a viewer, I want to open the reports I was sent without signing up, so I can help right away.

Acceptance criteria

- Opening a link shows the shared reports on the model, with their findings, scan images, and original reports, and the viewer can step through and compare them in exam order
- Nothing about the patient beyond the shared reports is shown, including their email address
- A viewer cannot add, change, delete, or re-share anything
- An expired or revoked link says so and shows nothing else
    - If a link led to a copy and the original was deleted, the copy would be considered "expired."

F11: Trying Without an Account

S11.1: Trying the app as a guest

As a guest, I want to add one report without creating an account, so I can see whether the app helps me before giving it my email address.

Acceptance criteria

- Before adding a report, a guest is told it will be deleted within 24 hours
- A guest can add one report and use everything in F2, F3, and F4 for it
- A guest cannot add a second report, use the timeline, attach scan images, or share
- A guest's report and everything made from it is deleted within 24 hours of being added, and can be reached only from the browser it was added in

S11.2: Keeping a guest report

As a guest, I want to keep my report when I create an account, so I don't have to add it and check it again.

Acceptance criteria

Given I added a report as a guest less than 24 hours ago
When I create an account or sign in, in the same browser
Then the report and its checked findings move into that account

6. Non-Functional Requirements

These twelve requirements apply across every feature, and each one can be tested and failed.

ID	Requirement	Definition
NFR1	Time to findings	The findings for a report of up to 10 pages are ready to check within 10 seconds of adding it.
NFR2	Time to model	Once findings are confirmed, the model appears with every marker in place within 5 seconds.
NFR3	Smooth movement	Turning and zooming the model runs at 30 frames per second or more on a laptop with built-in graphics made in 2020 or later.
NFR4	Report integrity	The original report and attached images are always shown exactly as added. Nothing the app produces changes them or is mixed into them, and the app's own content is always visibly separate from the report's.
NFR5	Honest model	Every view of the model says it is a standard skeleton, not the patient's own bones. Every marker traces to a finding from the report or one the patient entered; nothing is placed by guesswork.

ID	Requirement	Definition
m	No machine-written medical text	Every bone name, explanation, and status the app shows was written in advance and checked by a teammate. No medical text, including any summary, is produced by software while the app runs; this rules out AI text generation.
NFR7	Reading accuracy	On the team's test set of reports (OQ1), before any correction by the patient, at least 90% of stated fractures are found with the correct bone and side, and no more than 5% of findings name a bone or side the report does not.
NFR8	Personal data	The app asks for nothing about a person beyond their email address and password. A report and everything made from it can be seen only by its account holder and through links they create or users they give direct access to, and is used for nothing else.
NFR9	Link privacy	A share link cannot be found by browsing, searching, or guessing, and opens only the reports it covers. Reports that are shared directly (account → account/email, rather than via links) follow similar logic.
NFR10	Deletion	From the moment a report or account is deleted, no one can reach anything in it, and every stored copy is erased within 30 days.

ID	Requirement	Definition
NFR11	First use	In a test with at least 5 people who have never used the app, at least 4 add a sample report and find its fracture on the model within 3 minutes, without help.
NFR12	Legibility	Marker labels, the findings list, and report text are readable on a 13-inch laptop screen without zooming the browser. No information is shown by color alone.

7. Open Questions

Two decisions are still open. Each names what we are waiting on, who decides, and what it blocks.

OQ2: Can text be read reliably from photos of printed reports?

Screenshots and scanned PDFs are clean images and usually read well. Phone photos of paper are taken at angles, in poor light, and with shadows, and one misread word can move a fracture to the wrong bone. The text check in S8.1 catches some errors, but not text too poor to correct.

- **Waiting on:** A test with 10 phone photos of printed sample reports, counting how many words per page need correcting
- **Decided by:** Team
- **Blocks:** F8. If photos fail, F8 covers screenshots and scanned PDFs only.

OQ3: When is a fracture in a later report the same injury?

Comparison (S5.2) and healing status (S5.3) match findings across reports. The simplest rule, same bone and same side, fails when a patient has two separate fractures of one bone, or a new injury months later. Radiologists often refer back with phrases like "again seen" or "previously described", which may be a better signal.

- **Waiting on:** Follow-up reports from the OQ1 test set, to see how often each case occurs
- **Decided by:** Team
- **Blocks:** F5. Until it is decided, S5.2 uses same bone and same side.

– Answered Questions –

OQ1: What reports do we build and test against?

F2 is only as good as the reports it is tested on, and NFR7 cannot be measured without a test set. Real reports contain personal health information, and few public collections include fracture reports. The

options are team members' or volunteers' own reports with personal details removed, example reports from public radiology teaching sites, or reports the team writes from published templates.

- **Answer:** The instructor allowed de-identified real reports. The test set is at least 30 fracture reports from team members and volunteers, with personal details removed, and any shortfall is written from Radiopaedia examples (Project Plan, Milestone 2).
- **Decided by:** Instructor and team
- **Blocks:** F2, F5, NFR7

8. Future Directions

These are left out of this iteration but are plausible next steps none of them is a commitment.

FD1: A model of the patient's own bones, from CT scans. A CT scan is a stack of hundreds of cross-sectional images, and the bone in it can be separated out and built into a true 3D model of the patient's skeleton. This is the only practical path to showing a patient's real bones, because a single X-ray image has no depth (Appendix A.2). It needs the full scan files (DICOM), which patients must request from the hospital, and it is the largest item here.

FD2: Children's skeletons. A child's skeleton with growth plates, for fractures only children's bones have, such as greenstick, buckle, and growth-plate fractures. Today every report is shown on the standard adult skeleton (S3.1), including reports a parent adds for a child.

FD3: Spine, ribs, and skull. Vertebrae and ribs are numbered (C1 to L5, and ribs 1 to 12 on each side), and spine reports describe many findings that are not fractures. This needs its own term list and placement rules.

FD4: Phones and tablets. The same app on smaller screens, with touch controls for the model.

FD5: Caretaker accounts. Accounts linked with the patient's consent, so a caretaker can add reports on the patient's behalf and see new ones without a separate link for each. It would also let one account hold reports for several people, such as a parent with two children.

FD6: Translating the app. The app's own text and the term explanations in languages other than English. Reports themselves would still need to be written in English (Section 2.2).

FD7: Surgical hardware and casts. Plates, screws, rods, and casts mentioned in a report, drawn on the model.

FD8: Reminders. Reminders for follow-up appointments and the next X-ray.

FD9: Importing from patient portals. Pulling reports straight from a hospital's patient portal, so nothing has to be downloaded and added by hand.

9. References

Document	Covers
Software Pitch	Why we are building this, and for whom

Document	Covers
Requirements Document, version 1	The team's first draft, which this version replaces
Radiopaedia	Public radiology reference for fracture terms and report wording, used to build the term list (F9) and Appendix A
Merck Manual Skeletal System Z-Anatomy	Potential inspiration for the interactive 3D model system used in the application

Appendix A: Reading an Imaging Report

Nothing here is a requirement. It is what a developer needs to know to build F2 and F3 without guessing.

A.1 The parts of a report

Most imaging reports follow the same order, and each part matters differently to the app.

Section	What it says	Why it matters here
Exam	What was imaged, which side, how many views, and when (the date may also sit in the header)	Exam date (S1.2), side when a finding gives none (S2.1), views (S7.2)
History or Indication	Why the exam was ordered, such as "fall on outstretched hand"	Not a finding; never marked
Comparison	Earlier exams this one is compared with, and their dates	Never the exam date (S1.2)
Technique	How the images were taken	Ignored
Findings	Detailed observations, bone by bone	The main source of findings
Impression	The radiologist's conclusion, usually repeating the key findings	A fracture here and in Findings is one finding, not two (S2.1)

A.2 Types of imaging

- **X-ray (radiograph).** One flat image of a body part, usually taken from two or three angles. Bones overlap and there is no depth information, so a single X-ray image cannot be turned into an accurate 3D model of the patient's bone. Most fractures are diagnosed and followed up with X-rays.
- **CT.** Hundreds of thin cross-sectional images that stack into a 3D volume. A model of the patient's own bone can be built from a CT (FD1).
- **MRI.** Cross-sectional images that show soft tissue well, often used to find fractures an X-ray misses.
- **DICOM.** The file format for full scan files. A JPEG, PNG, or screenshot of a scan is a flat copy of one image and loses the rest of the scan.

A.3 Which bones the app covers

The app covers the appendicular skeleton: the bones of the arms and legs, plus the shoulder and hip bones that attach them, 126 bones in all.

Region	Bones	Count (both sides)
Shoulder	Clavicle (collarbone), scapula (shoulder blade)	4
Arm and forearm	Humerus, radius, ulna	6
Wrist (carpals)	Scaphoid, lunate, triquetrum, pisiform, trapezium, trapezoid, capitate, hamate	16
Hand	5 metacarpals and 14 phalanges per hand	38
Pelvis	Hip bone, made of the ilium, ischium, and pubis	2
Thigh, knee, and leg	Femur, patella (kneecap), tibia, fibula	8
Ankle and midfoot (tarsals)	Talus, calcaneus (heel bone), navicular, cuboid, and the medial, intermediate, and lateral cuneiforms	14
Foot	5 metatarsals and 14 phalanges per foot	38

Reports often use adjective forms: radial (radius), ulnar (ulna), humeral (humerus), femoral (femur), tibial (tibia), fibular (fibula), and calcaneal (calcaneus). The app does not cover the skull, the spine (including the sacrum and tailbone), the ribs, the breastbone, or the hyoid. Those 80 bones are the axial skeleton (FD3).

A.4 Directions and positions

- **Anatomical position** is the reference pose for every direction word: standing, facing forward, arms at the sides, palms forward.
- **Left and right** always mean the patient's own. In a front view, the patient's left is on the viewer's right. Getting this backwards puts every marker on the mirror-image bone, and it is the easiest mistake to make in this app.
- **Proximal and distal** mean nearer to and farther from the center of the body. The distal radius is its wrist end; the proximal humerus is its shoulder end.
- **Parts of a long bone:** the shaft is the middle, the metaphysis is the flared part near each end, and named parts include the head, neck, base, and condyles.
- **Medial and lateral** mean toward and away from the body's midline. In anatomical position the radius is on the thumb side (lateral) and the ulna on the little-finger side (medial).
- **Named spots:** the medial malleolus is the inner ankle bump, at the end of the tibia; the lateral malleolus is the outer bump, at the end of the fibula. Many hip fractures are in the neck of the femur.

- **Fingers and toes** are numbered 1 to 5, starting from the thumb or big toe. Reports use names or numbers: the fifth metacarpal is the hand bone leading to the little finger. The thumb and big toe have two phalanges each, with no middle phalanx; the other digits have three.

A.5 Fracture words

These words describe the break. Each needs a plain-language explanation for F9, checked by a teammate.

Term	Meaning
Nondisplaced	The broken pieces are still lined up
Displaced	The pieces have moved out of line
Angulated	The pieces meet at an angle instead of in a straight line
Comminuted	Broken into more than two pieces
Transverse, oblique, spiral	Broken straight across, at a slant, or twisting around the bone
Impacted	The pieces are pushed into each other
Avulsion	A small piece pulled off where a tendon or ligament attaches
Intra-articular	The break reaches into a joint
Stress (hairline)	A thin crack from repeated strain
Occult	Not visible on the X-ray; seen on another scan, or suspected

Named fractures stand for a specific bone and part (S2.1):

Name	Bone and part
Colles fracture	Distal radius, tilted toward the back of the hand
Smith fracture	Distal radius, tilted toward the palm
Scaphoid fracture	Scaphoid, in the wrist
Boxer's fracture	Neck of the fifth metacarpal
Hip fracture	Top of the femur: the neck, or just below it
Jones fracture	Fifth metatarsal, near its base

A.6 Certainty and healing words

Kind	Typical words	Used in
Absent	no fracture, negative for fracture, no acute fracture	No finding (S2.2)
Uncertain	possible, probable, suspected, questionable, cannot be excluded	Uncertain finding (S2.2)
New	acute	Healing status (S5.3)
Healing	healing, callus formation, interval healing	Healing status (S5.3)
Healed	healed, united, remodeled, old	Healing status (S5.3)
Not healing	nonunion, delayed union	Healing status (S5.3)
Healed out of line	malunion	Healing status (S5.3)
Alignment	alignment unchanged, increased displacement	Recorded with the finding (S2.1)
Refers back	again seen, previously described, compared with an earlier date	Matching across reports (OQ3)

A.7 X-ray views

The Exam line usually says how many views were taken and sometimes names them, as in "left wrist, 3 views".

View	Taken from	Model view (S7.2)
Frontal (AP or PA)	Front to back (AP) or back to front (PA); the two look almost the same	Front
Lateral	The side	Side
Oblique	An angle, usually about 45 degrees	Front, turned about 45 degrees

A.8 Where patients get reports

- Patient portals often release imaging reports as soon as they are final, mostly before the ordering doctor has gone over them with the patient.
- Portals usually show the report as on-screen text, as a PDF to download, or both.

- Imaging centers can provide a printed copy, and a disc or download of the full scan files (DICOM), often with a basic viewer.

Appendix B: Worked Example

These two reports are fictional, written by the team in standard report style. They show what F2 and F5 must produce, and can serve as the first acceptance test.

B.1 First report

```
EXAM: XR WRIST LEFT, 3 VIEWS
DATE OF EXAM: 09/14/2026
HISTORY: Fall on outstretched hand.
COMPARISON: None.
FINDINGS: There is a transverse fracture of the distal radial metaphysis with
mild dorsal angulation. No fracture of the ulna. The carpal bones are intact.
Soft tissue swelling over the dorsal wrist.
IMPRESSION: Mildly angulated distal left radius fracture.
```

Bone	Side	Part	Words recorded	Certainty	Healing status
Radius	Left	Distal, metaphysis	transverse; mild dorsal angulation	Certain	None stated

- The exam date is September 14, 2026 (S1.2).
- The fracture appears in both Findings and Impression, but it is one finding (S2.1).
- The Findings sentence names no side, so the finding takes "left" from the Exam line (S2.1).
- "No fracture of the ulna" creates no finding (S2.2).
- Soft tissue swelling is not a fracture and is not marked (Section 2.2).
- "3 views" names no views, so attached images are labeled front, side, or angled (S7.2).

B.2 Follow-up report

```
EXAM: XR WRIST LEFT, 2 VIEWS
DATE OF EXAM: 10/12/2026
HISTORY: Follow-up of distal radius fracture.
COMPARISON: 09/14/2026.
FINDINGS: Again seen is the distal radius fracture with interval callus
formation. Alignment is unchanged. No new fracture.
IMPRESSION: Healing left distal radius fracture. Alignment unchanged.
```

Bone	Side	Part	Words recorded	Certainty	Healing status
Radius	Left	Distal	alignment unchanged	Certain	"interval callus formation", "healing"

- The exam date is October 12, 2026; the Comparison date, September 14, 2026, is not used (S1.2).
- The History line mentions the fracture but is not a finding (Appendix A.1).

- "No new fracture" creates no finding (S2.2).
- The finding is on the same bone and side as in the first report, and "again seen" refers back, so it is the same injury (S5.2, OQ3).
- Comparing the two reports shows the fracture in both, with "healing" as its status in the later one (S5.2, S5.3).

Appendix C: Skeleton 3D Model Sourcing and Implementation Plan

This section outlines the planned 3D asset sourcing and implementation for the web application.

C.1 Source

- The model would come from an open-source distributor to maximize development time on the project
    - Likely source: z-anatomy.com; GitHub repository

C.2 Model Structure

- The model would contain all 126 bones of the appendicular skeleton (A.3).
- The model would be segmented into 126 meshes, each representing a bone. This allows the bones to be accessed efficiently.
- The model would be under 250,000 triangles (total across all meshes).